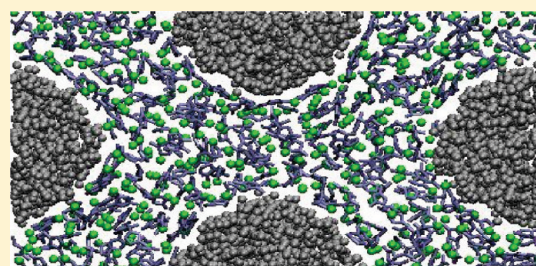


Effects of Pore Size and Pore Loading on the Properties of Ionic Liquids Confined Inside Nanoporous CMK-3 Carbon Materials

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ABSTRACT: We report molecular dynamics simulations of the structural and dynamical properties of 1,3-dimethylimidazolium chloride, $[\text{dmim}^+][\text{Cl}^-]$, confined inside the ordered mesoporous carbon CMK-3. This material exhibits interconnected nanopores with heterogeneities in pore shape, pore size, and pore surface roughness. Our results indicate that variables such as pore size, pore loading, and pore morphology have a profound influence on the structural and dynamical properties of the confined IL. Significant layering is observed in all the systems, with the number of layers and the relative positions of the density peaks of the ions varying with pore size and pore loading. At low pore loadings, the density profile along the axial direction becomes highly heterogeneous, indicating the presence of regions where $\rho \approx \rho_{\text{bulk}}$ and regions partially depleted of IL for which $\rho \ll \rho_{\text{bulk}}$. The radial distribution functions suggest that the structure of the confined IL is similar to that of a bulk IL; however, important variations in the height of the peaks are observed as the pore loading changes. Our results indicate that the dynamics of the confined IL are significantly slower than those observed in bulk systems. The mean squared displacements (MSDs) of the confined ions in the three directions are of similar magnitude, in contrast to what was observed when the same IL is confined inside a slit-like pore of similar size. For pore loadings similar to ρ_{bulk} , the MSDs in the axial direction increase monotonically with pore size. For fixed pore sizes, the axial MSDs decrease monotonically as the pore loading increases above ρ_{bulk} but below ρ_{bulk} we observe nonmonotonic variations in the axial MSDs. Our results indicate that, for any given pore size, the axial diffusivity of the confined cations reach maxima at pore loadings below ρ_{bulk} , possibly due to the large heterogeneities observed in the axial density profile at these low pore loadings.



1. INTRODUCTION

A fundamental understanding of the behavior of ionic liquids (ILs) confined in materials with nanometer-sized pores is relevant for their application as alternative electrolytes for energy storage in electrochemical double-layer capacitors (EDLC)^{1–12} and for conversion of solar energy in dye-sensitized solar cells (DSSCs).^{13–19} ILs can allow EDLCs to operate at higher voltages, boosting their specific energy and specific power. ILs also are nonvolatile and nonflammable and exhibit larger electrochemical windows as compared to ‘conventional’ electrolytes (i.e., aqueous or organic solutions of ‘small’ ions). The macroscopic performance of IL-based EDLCs and DSSCs is governed by the properties of the electrical double layer at the IL–nanoporous electrode interface, which in turn are determined by the structural and dynamical properties of the IL confined inside the nanopores. For example, recent experimental and theoretical investigations for conventional electrolytes suggest that controlling the nanostructure of porous carbons can lead to increases of up to 50% in the storage capacity of EDLCs.^{20–22} Nevertheless, the fundamental understanding of the properties of the electrical double layer at the IL–nanopore interface is still in its early stages.²³ In particular, molecular simulations can provide useful insights and complement experimental efforts in this area. A few molecular simulation studies have been published on ILs confined inside nanopores of

simple geometry, such as slit-like^{24–29} and cylindrical.^{30–33} These studies have provided important insights about the properties of ILs confined inside pores of simple geometries. Nevertheless, simple pore models typically neglect the effects of factors such as heterogeneities in pore shape and size, pore surface roughness, and surface chemistry, which may be present in real nanoporous materials.

Here we report molecular dynamics (MD) simulations of the IL 1,3-dimethylimidazolium chloride, $[\text{dmim}^+][\text{Cl}^-]$, confined in a realistic model³⁴ of the ordered mesoporous carbon (OMC) CMK-3.^{35–38} OMCs are typically synthesized through a ‘nanocasting’ procedure,^{36–38} which involves (1) infiltration of a carbon precursor into a hard template (in the case of CMK-3, the template is a mesoporous silica, SBA-15,^{39,40} which exhibits hexagonally ordered cylindrical pores); (2) conversion of the precursor into a solid via polymerization and carbonization; and (3) removal of the template by dissolution. The structure of the resulting carbon material is thus determined by the silica template. Therefore, CMK-3 consists of nanorods with uniform diameters, which are made of amorphous carbons, and their walls have corrugations

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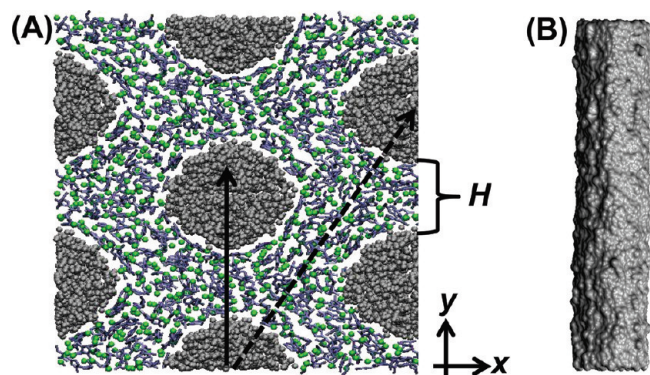


Figure 1. (A) Representative snapshot of the IL $[\text{dmim}^+][\text{Cl}^-]$ confined in a model CMK-3 material with a pore size of 1.8 nm. The cations and anions are depicted in purple and green. The solid and dashed lines show the paths used in our analysis (see text). The simulation box was replicated once in the y direction in this figure. (B) Close view of a model carbon rod in CMK-3, showing surface roughness at atomic length scales.

and curvature. The nanorods in CMK-3 are packed in a hexagonal pattern, and thus, the nanopores are interconnected in a regular way (Figure 1). OMCs are promising candidates for applications in EDLCs due to their controlled and narrow pore size distributions, large specific surface areas, and regularly interconnected pore network, which produce a lower resistance to ion transport. Different ordered mesoporous carbons have been used in EDLCs in recent studies (for recent reviews see, e.g., refs 7, 8, and 12); however, improvements in performance are still needed.^{7,8,12} Furthermore, the fact that OMCs like CMK-3 exhibit interconnected pores with limited heterogeneities in pore size, pore shape, and surface roughness make these materials well suited for fundamental studies aimed at determining the effect of such heterogeneities on the structural and dynamical properties of confined ILs. The IL we have chosen for our study, $[\text{dmim}^+][\text{Cl}^-]$, has been widely used in previous simulation reports.^{24,25,27,28,41–44} We focus our study on the effects of variations in pore size and pore loading on the structure and dynamics of $[\text{dmim}^+][\text{Cl}^-]$ inside the model CMK-3 material by analyzing the density profile, molecular orientation, and mobility of the IL. The remainder of the paper is structured as follows. Section 2 introduces the computational models and methods used in this study. In section 3 we present and discuss the structural and dynamical properties of the confined IL for different pore sizes and pore loadings. Finally, in section 4 we summarize our findings.

2. COMPUTATIONAL DETAILS

Classical MD simulations have been carried out with the IL 1,3-dimethylimidazolium chloride, $[\text{dmim}^+][\text{Cl}^-]$, confined inside a model CMK-3 material at a temperature of 425 K. All simulations were performed using the GROMACS MD package.⁴⁵ The IL was modeled using the explicit all-atom, nonpolarizable force field model developed by Lopes et al.,^{46–50} which was based on the optimized potential for liquids simulation/all atom (OPLS-AA) developed by Jorgensen et al.⁵¹ The force field of Lopes et al.^{46–50} is known to give dynamics that are slower than experimental values for the particular case of the IL $[\text{bmim}^+][\text{PF}_6^-]$ in the bulk.^{52,53} In addition, it has been argued that polarizable models of ILs are better than nonpolarizable models in reproducing the dynamics of pure ILs.^{54–57} However, polarizable models

Table 1. Systems Details

pore size	no. of IL pairs	dimensions (nm)	pore loading range
1.8 nm	1027–1412	$7.86 \times 4.54 \times 10.17$	80–110%
1.2 nm	640–880	$6.82 \times 3.94 \times 10.17$	80–110%
0.9 nm	398–96	$6.30 \times 3.64 \times 10.17$	65–125%
1.2 nm ^a	177	$5.72 \times 5.60 \times 3.30$	100%

^aSlit pore model.

are more complex and computationally more expensive; furthermore, it has been argued that simulations of ILs using nonpolarizable and polarizable force fields give similar dynamics at high temperatures.⁵⁷ We also decided to use the force field of Lopes et al.^{46–50} because it has been extensively used in previous simulation studies, including confined systems.^{27,28,30–32} The initial atomic positions of the cation were generated using PRODRG.⁵⁸ Prior to studying confined systems, we carried out MD simulations of bulk $[\text{dmim}^+][\text{Cl}^-]$ in the NPT ensemble. From these simulations we obtained a bulk density of 1.132 g/cm^3 , which is less than 1% difference from the experimental density found by Fannin.⁵⁹ The $g(r)$ for cation–anion and anion–anion in the bulk were found to be in excellent agreement with similar results reported by Sha et al.²⁷ Furthermore, we also carried out preliminary simulations of $[\text{dmim}^+][\text{Cl}^-]$ confined in a slit pore of size $H = 0.9 \text{ nm}$, and the cation–anion and anion–anion $g(r)$ were in excellent agreement with those determined by Sha et al.²⁷

The model CMK-3 material we used in this study was developed by Jain and co-workers.³⁴ As mentioned above, CMK-3 consists of carbon nanorods with uniform diameters that are arranged in a honeycomb pattern. To obtain one model carbon rod, Jain et al. performed grand canonical Monte Carlo (GCMC) simulations to model the adsorption of a ‘fictitious’ monatomic ideal gas of carbon inside a cylindrical silica pore mimicking a SBA-15 material. The resulting carbon rod had a length of 10.17 nm and an average diameter of $\sim 2.8 \text{ nm}$ (determined as the average center-to-center distance between diametrically opposed carbon atoms in the rods, plus the LJ size parameter of a carbon atom). The carbon rod was then duplicated and arranged in a simulation box arranged in a hexagonal pattern (Figure 1). Further details of this model CMK-3 material can be found elsewhere.³⁴ Recently, Peng et al. studied H_2 storage in a model CMK-5 material (this material is similar to CMK-3 but consists of carbon nanopipes instead of nanorods).⁶⁰ In their study, Peng et al.⁶⁰ modeled the carbon nanopipes in CMK-5 as several structureless, coaxial layers of cylindrically rolled graphene sheets. In contrast, the carbon rods in the model CMK-3 material of Jain et al.³⁴ consist of amorphous carbon and exhibit surface roughness at atomic scales (Figure 1).

For the simulations of confined IL, the carbon atoms in the CMK-3 material were modeled as LJ spheres with $\sigma_c = 0.340 \text{ nm}$ and $\epsilon_c/k_B = 28.0 \text{ K}$. All MD simulations for confined systems were carried out in the canonical (constant NVT) ensemble (Figure 1 and Table 1). The cutoff of Lennard–Jones interaction was taken as $10 \text{ \AA}$. The long-range coulomb interactions were handled by particle mesh Ewald (PME)⁶¹ with a cutoff of $12 \text{ \AA}$. In each simulation, the hydrogen bond lengths were constrained with the LINCS algorithm⁶² and a time step of 1 fs was used. Periodic boundary conditions were applied in all directions. The volume accessible to the IL was determined as the volume of the box minus the volume of the two carbon rods. Likewise, the pore size was defined as the minimum surface-to-surface distance between carbon rods (i.e., the distance between the centers of the

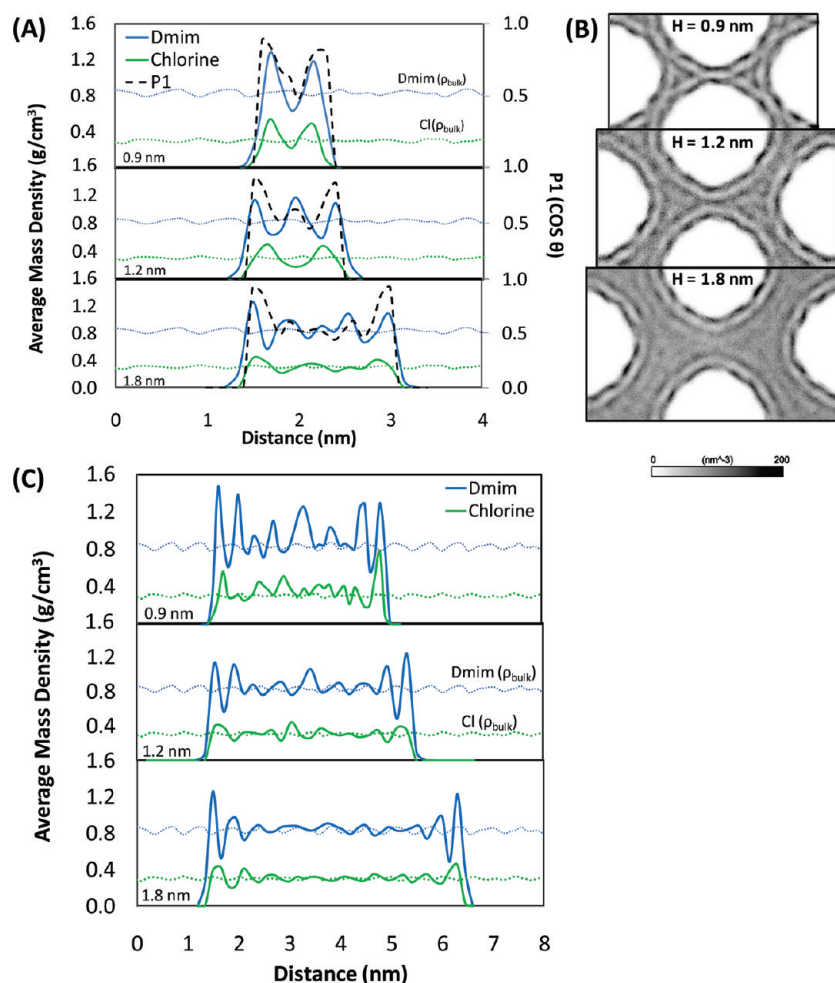


Figure 2. (A) Mass density profile of the confined ions (left axis) and orientation of the cations (right axis) along the solid-line path in Figure 1. (B) Density map of the cations for the three pore sizes considered in this study. (C) Mass density profile of the confined ions along the dashed-line path in Figure 1. In all cases, the confined IL has a density similar to ρ_{bulk} . The dotted lines represent the density profiles of the ions in a bulk system.

two nanorods minus the 2.8 nm diameter). It is important to note that the pore size is approximated, as the carbon rods exhibit surface roughness and their diameter is not completely uniform. The carbon atoms in the nanorods were kept in fixed positions throughout our simulations. In this study we ran simulations for three pore sizes, 0.9, 1.2, and 1.8 nm. The pores of these CMK-3 materials were filled with varying loading amounts of the IL, expressed as a percentage of the IL bulk density at 425 K (ρ_{bulk}). The range of pore loadings considered here varied between 80% and 110% of ρ_{bulk} for the 1.2 and 1.8 nm pore sizes and between 65% and 125% for the 0.9 nm pore size. The dimensions and number of ion pairs per system are presented in Table 1. For comparison purposes, we also performed calculations of the same IL confined inside a slit-like graphitic pore with a pore size of 1.2 nm (Table 1). The improved velocity-rescaling algorithm recently proposed by Parrinello et al.^{63,64} was used to mimic weak coupling at 300 K with a coupling constant of 0.1 ps.

In our simulations the ions were initially placed in an arbitrary lattice inside the model CMK-3 material. We first minimized the energy of our initial configurations using the steepest descent method. Afterward, the systems were melted at 600 K for 3 ns and then annealed from 600 to 425 K in three stages: 1 ns at 600 K, 1 ns at 500 K, and 1 ns at 425 K. Our systems were further relaxed at 425 K for 4 ns and averages accumulated over a minimum

of 10 ns. Our results for the mean square displacements of the ions (Figures 7 and 8) suggest that these simulation times are long enough so that the Fickian (diffusive) regime is reached and properly sampled. Furthermore, results for the single-particle time correlation functions for the rotational motion of the cations (Figure 6) also suggest that these simulation times are long enough for the rotational motion to decorrelate. To further attempt to overcome the difficulties posed by the slow dynamics of confined ionic liquids (which are likely to be exacerbated when these compounds are confined inside nanopores), for each pore size we conducted a second set of simulation runs starting from different initial configurations (by taking the last configuration of our first set of runs and repeating the steps of melting, annealing, relaxing, and averaging described above). Therefore, for each pore size, the results presented in this paper were averaged over two different realizations of the same system.

3. RESULTS

Structural Properties. In Figure 2 we present the mass density profiles of the ions (in g/cm³) confined inside model CMK-3 carbons with three pore sizes, 0.9, 1.2, and 1.8 nm. In all cases the density of the confined IL is $\rho \approx \rho_{\text{bulk}} = 1.132 \text{ g/cm}^3$. The density profiles presented in Figure 2A were determined

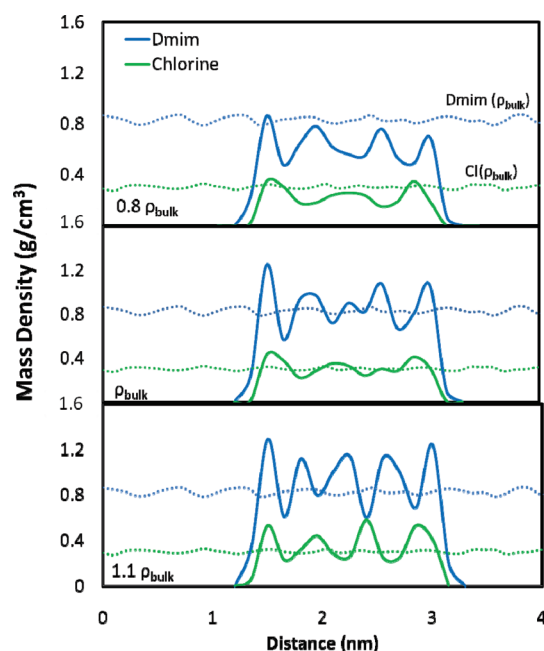


Figure 3. Mass density profile of the confined ions along the solid-line path in Figure 1 for different pore loadings. In all cases, the model CMK-3 material has a pore size of 1.8 nm. The dotted lines represent the density profiles of the ions in a bulk system.

along the straight-line path indicated in Figure 1, whereas the density profiles shown in Figure 2C were determined along the dashed-line path depicted in Figure 1. The density profile of a simulated bulk system, $\rho \approx \rho_{\text{bulk}}$, was also included as a reference. In Figure 2A the mass density profiles exhibit oscillatory behavior, indicating significant layering. The maxima in the local density are observed close to the surfaces of the carbon nanorods for both ions. The number of layers of ions along the straight-line path (Figure 1) increases from 2 to 3–4 as the pore size increases from 0.9 to 1.8 nm. Further evidence of layering can be observed in the density maps for the cation shown in Figure 2B. The position of the density peaks for $[\text{dmim}^+]$ and $[\text{Cl}^-]$ also depends strongly on pore size. For a pore size of 0.9 nm (Figure 2A), the peaks for $[\text{dmim}^+]$ and $[\text{Cl}^-]$ are observed at similar values of distance for this particular pore loading ($\rho \approx \rho_{\text{bulk}}$). In contrast, for a pore size of 1.8 nm (Figure 2A), an increase in available space allows the anions to position themselves between the cation layers.

To further understand the structural properties of the ions, we also determined the orientation of the cations using the Legendre polynomial $P_1(\cos(\theta))$, where θ is defined as the angle between the normal of the imidazolium ring and the surface normal.^{65,66} The average orientation of the ions is represented by the dashed lines in Figure 2A. The local maxima near the rod surfaces show that the imidazolium rings orient parallel to the surface, in analogy to what was observed in previous simulations studies of ILs confined in slit-like and cylindrical carbon pores.^{25,27,30,32}

In Figure 2C we present density profiles along the dashed line in Figure 1. In analogy to the results shown in Figure 2A, the maxima in the density profiles for the ions are again observed near the surface of the carbon rods. However, as the pore size increases it can be observed that the layering effects tend to diminish away from the nanorod surfaces and the density of the ions decrease to values similar to those observed in nonconfined systems.⁶⁷ The two different cross sections of the density depicted

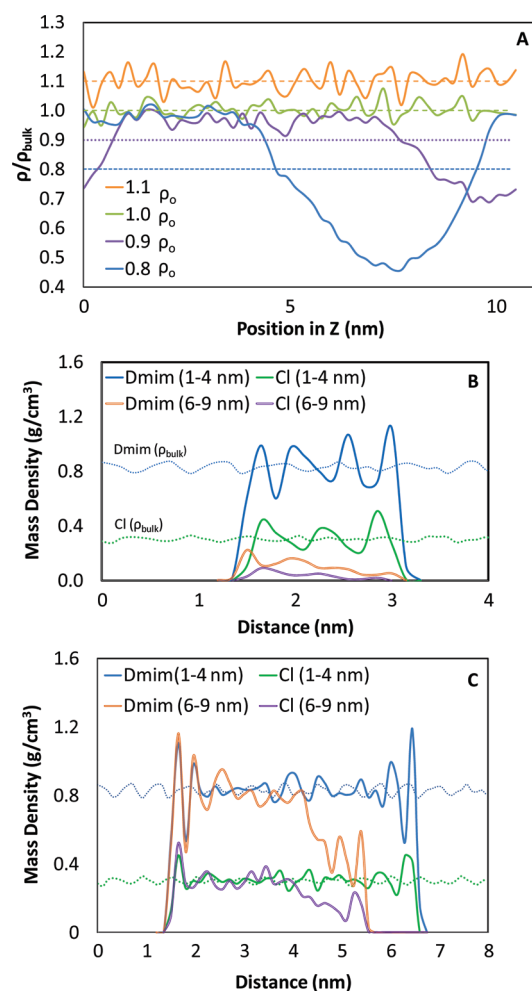


Figure 4. (A) Average mass density profile (normalized with respect to the bulk density ρ_{bulk}) along the axial direction for the IL confined inside a CMK-3 material with a pore size of 1.8 nm and different pore loadings. The 'nominal' density of the system for the different pore loadings is depicted as dashed lines. (B) Mass density profiles of the confined ions for a pore loading $\rho = 0.8 \rho_{\text{bulk}}$ along the solid-line path in Figure 1, averaged over different values of the axial z coordinate: $z = 1-4$ and $6-9$ nm. The dotted lines represent the density profiles of the ions in a simulated bulk system. (C) Similar to B but the density profiles were determined along the dashed-line path in Figure 1.

in Figure 2A and 2C (density profiles along the continuous and dashed lines in Figure 1, respectively) indicate that the pore geometry of CMK-3 causes the layering behavior to be anisotropic. As a result, the layering behavior in a CMK-3 material is expected to be significantly different from those observed in simple slit-like and cylindrical pores.

In order to study the effects of pore loading on the structural properties of the confined $[\text{dmim}^+][\text{Cl}^-]$, in Figure 3 we present results for the mass density profiles observed along the solid line of Figure 1 for a CMK-3 material with a pore size of 1.8 nm and different pore loadings. In analogy to the results shown in Figure 2, the density profiles shown in Figure 3 exhibit local maxima near the surfaces of the carbon rods as well as significant layering effects. Similar to the bulk density, the pore loading does not affect the anions position between the cation peaks for the 1.8 nm pore size. However, additional layers of ions arise as pore loading increases, which is an effect that is also observed for the other

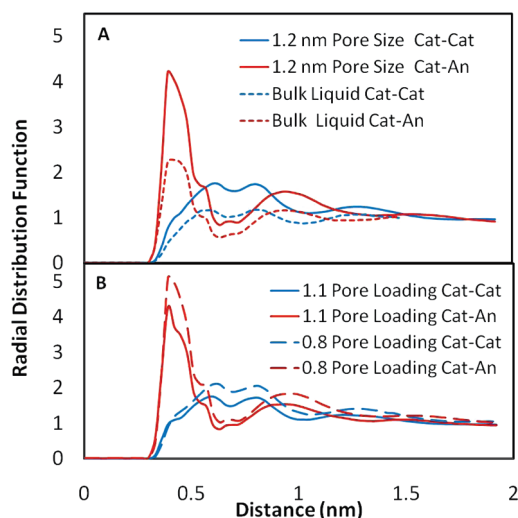


Figure 5. (A) Comparison of radial distribution functions of the bulk IL with those of an IL confined inside a model CMK-3 material with a pore size of 1.2 nm. (B) Comparison of radial distribution functions of the confined IL at different pore loadings and a pore size of 1.2 nm.

pore sizes considered in this study. It is important to note that in Figure 3 the densities were averaged over the whole axial direction. For low pore loadings ($\rho < \rho_{\text{bulk}}$), the average density profile (as shown in Figure 3 for $\rho = 0.8\rho_{\text{bulk}}$) may not be representative of the local structure of the system (see Figure 4 and its discussion in the next paragraph).

In Figure 4A we present results for the density profile in the axial direction for $[\text{dmim}^+][\text{Cl}^-]$ confined in a CMK-3 material with a pore size of 1.8 nm and different pore loadings. In this figure, for any given value of the axial z coordinate, we plotted the density of the IL averaged over the whole cross section and then divided it by the bulk density ($\rho = 1.132 \text{ g/cm}^3$). The 'nominal' density of the system is depicted as dashed lines of different colors for the different pore loadings. At high pore loadings, the axial local density fluctuates slightly around the average density of the system. In contrast, at lower pore loadings ($\rho < \rho_{\text{bulk}}$), the density profiles in the axial direction exhibit large fluctuations. These results suggest that the ions prefer to cluster together, creating regions where the local density is similar to ρ_{bulk} and regions where $\rho \ll \rho_{\text{bulk}}$ and thus are partially depleted of IL. The data shown in Figure 4A suggests that for the largest pore loadings considered ($\rho = 1.1\rho_{\text{bulk}}$ and $\rho = \rho_{\text{bulk}}$) the fluctuations in the local density at different values of the axial z coordinate are small when compared to those observed at the lower pore loadings ($\rho = 0.9\rho_{\text{bulk}}$ and $\rho = 0.8\rho_{\text{bulk}}$). Therefore, for the largest pore loadings, the density profiles shown in Figures 2 and 3 (which were averaged over the axial direction) will not differ significantly from the individual density profiles measured at different values of the axial z coordinate. Regarding the lower pore loadings ($\rho = 0.9$ and $0.8\rho_{\text{bulk}}$), the density profiles measured along the continuum and dashed lines in Figure 1 will depend strongly on the value of the axial z coordinate and therefore may differ significantly from the density profiles averaged over the whole axial direction (e.g., Figure 3 for $\rho = 0.8\rho_{\text{bulk}}$). In Figure 4B and 4C we present the density profiles for a pore loading of $\rho = 0.8\rho_{\text{bulk}}$ along the continuum and dashed lines in Figure 1, averaged over different values of the z coordinate: $z = 1\text{--}4$ nm (representing the regions of high local density, see Figure 4A) and $z = 6\text{--}9$ nm (representing the regions with the lowest local density, see

Figure 4A). The data shown in Figure 4B and 4C suggests that for $z = 1\text{--}4$ nm the density profiles do not differ significantly from those presented in Figure 2A and 2C for $H = 1.8$ nm and $\rho = \rho_{\text{bulk}}$. In contrast, for $z = 6\text{--}9$ nm, the density profiles shown in Figure 4B are significantly smaller than those observed for $z = 1\text{--}4$ nm. The density profiles shown in Figure 4B are significantly different from those presented in Figure 3 for $\rho = 0.8\rho_{\text{bulk}}$, which were averaged over the whole axial direction. Similarly, in Figure 4C the profiles for $z = 6\text{--}9$ nm exhibit a significant drop in density at a distance of about 4 nm and eventually reach zero at a distance of about 5.5 nm. Similar trends are observed for the other pore sizes considered in this study. These heterogeneities in the axial density profiles may influence the dynamics at these low pore loadings (see next section).

In Figure 5 we present different radial distribution functions between the cations and anions confined in a CMK-3 material with a pore size of 1.2 nm and different pore loadings. Equivalent $g(r)$ functions for the bulk IL are also shown in this figure. For all pore loadings, the positions of the peaks and the features in the $g(r)$ of the confined IL are similar to those observed in the $g(r)$ of the bulk IL. However, the heights of the different peaks and shoulders tend to increase monotonically as the pore loading decreases. These results suggest that for all pore loadings considered in this study the confined IL has a liquid-like structure that is not significantly different from the structure of the IL in the bulk. It is important to note that the results presented in Figure 5 were averaged over the whole system. Therefore, the $g(r)$ results for low pore loadings ($\rho = 0.8\rho_{\text{bulk}}$) should be treated with caution, as the $g(r)$ averaged over the whole system may not be representative of the local structure of the system (due to the clustering effects discussed above in Figure 4). Similar results are observed for the other pore sizes considered in this study.

Dynamical Properties. The mobility of the confined ions in different directions is one of the factors determining the internal resistance in an EDLC, which ultimately affects its specific power.^{8–12} In Figure 6A the mean square displacement (MSD) is plotted as a function of time for $[\text{dmim}^+]$ and $[\text{Cl}^-]$ in the axial (z) and x directions when confined inside a CMK-3 material with a pore size of 1.2 nm. The density of the confined system is similar to ρ_{bulk} . Following the work of Urahata and Ribeiro,⁶⁸ in this figure we also plotted different single-particle time autocorrelation functions (ACF) for the rotational motion of the cations, namely, (1) the reorientation of the cation around an axis in the plane of the imidazolium ring, in the direction of a vector joining the N–N atoms (Cr^{NN}), and (2) similar to Cr^{NN} but now the rotation is in the direction of a vector joining the C–N atoms in the cation (Cr^{CN}). The ACF for the velocity of the center of mass of the cation was not included, as previous studies in bulk⁶⁸ and confined³⁰ systems suggest that the translational motion of the cation is orders of magnitude faster than its rotational motion. Our results indicate that about 8 ns are required for the rotational motion of the cations to decorrelate. These results contrast to what we found in our simulations of the same IL in the bulk, where the same rotational ACFs are decorrelated after about only 0.8 ns (results not shown). The time needed for the rotational motion of the cations to decorrelate increases as the pore size is reduced and the degree of confinement becomes larger (results not shown).

Regarding the MSD results shown in Figure 6A, our results indicate that the confined cations move faster than the anions, in analogy to previous findings.^{68–73} A second observation is the similarity in magnitude of the MSDs in the axial direction z (free

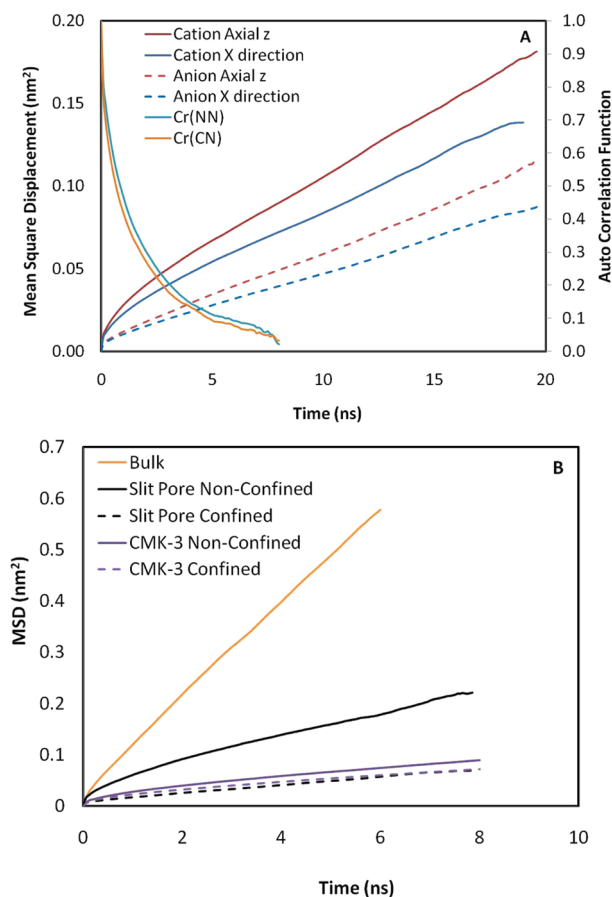


Figure 6. (A) Mean square displacement (left axis) in the axial (z) and x directions for the cations and anions confined in a CMK-3 material with a pore size of 1.2 nm and a pore loading similar to ρ_{bulk} . (Right axis) Single-particle time correlation functions for the rotational motion of the cations. The correlation functions depicted are the following: Cr^{NN} = reorientation of the cation around an axis in the plane of the imidazolium ring, in the direction of a vector joining the N–N atoms, and Cr^{CN} = similar to Cr^{NN} but now the rotation is in the direction of a vector joining the C–N atoms in the cation. (B) Mean square displacement in the confined (z) and nonconfined (x , y) directions for the cations and anions inside a graphitic slit-like pore with a pore size of 1.2 nm and a pore loading similar to ρ_{bulk} .

of confinement) and in the x and y directions (subject to confinement; $\text{MSD}(x) \approx \text{MSD}(y)$). These results are in contrast to what we obtained for the MSD of the same ions when confined inside a slit-like pore of similar size, $H = 1.2$ nm (Figure 6B). These results indicate that the displacement in the x and y directions (free of confinement; again, $\text{MSD}(x) \approx \text{MSD}(y)$) is much higher than the MSD values in the z direction (subject to confinement). These differences in dynamical behavior can be attributed to the larger degree of surface roughness of the carbon nanorods in CMK-3 (Figure 1), as compared to the relatively smooth walls in graphitic slit-like carbon nanopores. Differences in pore geometry can also explain the differences in the MSD of the ions. In a CMK-3 material, the hexagonal arrangement of the carbon rods and their surface curvature results in nonuniform confinement effects. In contrast, in a model slit pore, the distance between the pore walls is constant and confinement effects are uniform throughout the pore.

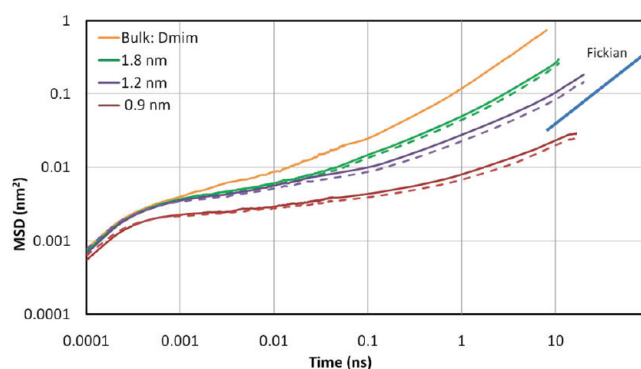


Figure 7. Mean square displacement of the confined cations in the axial (solid lines) and x directions (dashed lines). Results for three pore sizes and for a bulk system are compared. The continuous blue line indicates the behavior expected for the Fickian regime ($\text{MSD} \propto t$). In all cases, the pore loading is similar to ρ_{bulk} .

In Figure 7 we plot the MSD in the z (axial) and x directions for the IL confined in CMK-3 materials with different pore sizes, for a pore loading such that $\rho \approx \rho_{\text{bulk}}$. The MSD for the bulk ions is also depicted in Figure 7. The MSDs for the bulk ions are larger than those observed for the confined ions, indicating that the dynamics for bulk systems are faster than for confined systems. All MSDs exhibit the three dynamic regimes (ballistic, caged, and diffusive or Fickian).^{74,75} The ballistic regime occurs in the first few picoseconds when the molecules have not interacted with other neighbors and the dynamic relationship is much higher than the other regimes, $\text{MSD} \propto t^2$. In the caged regime the dynamics slow down due to the ions becoming temporarily trapped by their neighbors, and in the diffusive (Fickian) regime $\text{MSD} \propto t$. Our results indicate that the bulk ions reach the diffusive regime much faster than the confined ILs (<1 ns, as compared to several nanoseconds for the confined ions). As the pore size decreases, the dynamics of the confined ions remain in the caged regime for a longer period of time before finally reaching the diffusive regime. Other trends that are of interest in Figure 7 include a monotonic increase in the MSD values with increases in the pore size, with the maximum MSD found for the bulk system. Previous simulation studies on supercooled Lennard–Jones fluids inside nanopores,⁷⁴ and on the IL $[\text{bmim}^+][\text{PF}_6^-]$ confined inside carbon nanotubes³⁰ indicate that the molecules furthest from the surface experience the highest MSD values. As the pore size increases (and the diameter of the nanorods is kept constant), a larger fraction of the ions are positioned away from the surfaces and found to have larger MSDs. Therefore, the average MSD for the confined $[\text{dmim}^+][\text{Cl}^-]$ would increase monotonically with pore size, as observed in Figure 7. Nevertheless, our previous simulation results³⁰ for $[\text{bmim}^+][\text{PF}_6^-]$ inside carbon nanotubes show that the MSDs vary nonmonotonically with pore sizes. This apparent discrepancy in dynamic behavior could be caused by the important differences in pore geometry. Larger confinement effects are expected when a fluid is confined inside a cylindrical carbon nanotube, as compared to inside a CMK-3 material. Differences in the molecular structure of the ILs studied by us ($[\text{bmim}^+][\text{PF}_6^-]$ vs $[\text{dmim}^+][\text{Cl}^-]$) can also be a source of differences in dynamical behavior.^{52,73}

In the previous section we described the changes observed in the structural properties of the confined $[\text{dmim}^+][\text{Cl}^-]$ as the pore loading varied from $0.8 \rho_{\text{bulk}}$ to $1.1 \rho_{\text{bulk}}$ while keeping the pore size constant. In Figure 8 we compare the MSD in the axial

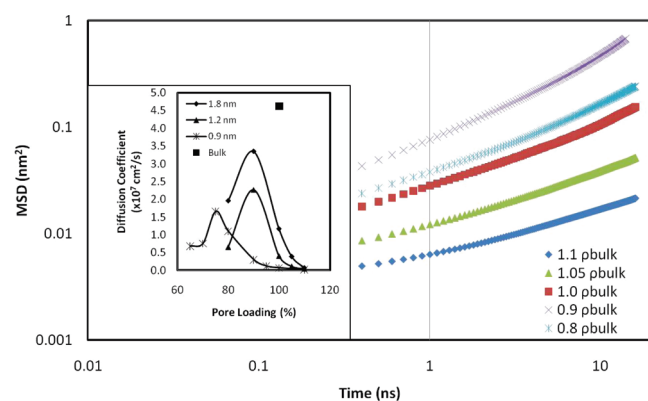


Figure 8. Mean square displacement of the cations in the axial direction as a function of pore loading, when they are confined inside a CMK-3 material with a pore size of 1.2 nm. (Inset) Diffusion coefficient of the confined cations in the axial direction for three different pore sizes and different pore loadings.

direction for the cations confined in a CMK-3 material with a pore size of 1.2 nm, as the pore loading is varied from 0.8 to 1.1 ρ_{bulk} . We do not show the MSDs for the anions or the MSDs for the ions in the x direction, because the results are consistent with the trends observed in Figure 6. Results from Figure 8 indicate that the axial MSD of the cations decrease monotonically as the pore loading increases above ρ_{bulk} . This trend can be explained by the reduced available volume and increased interactions between neighboring ions as the pore loading increases, which results in a reduction in the mobility of the ions. However, nonmonotonic variations in the axial MSDs are observed when the pore loadings vary below ρ_{bulk} for this pore size (1.2 nm), the largest MSD is observed for 0.9 ρ_{bulk} , followed by the MSD at 0.8 ρ_{bulk} and then the MSD at ρ_{bulk} . In the inset of Figure 8 we present results for the diffusion coefficient of the confined cations in the axial direction, as determined from the slope of the MSD in the diffusive regime, for different pore loadings and different pore sizes. We also include the calculated diffusivity of [dmim⁺] in the bulk, which is in good agreement with results from previous simulation studies using similar force fields.⁵² For all pore sizes and loadings, it was found that the axial diffusivity of the confined cations was always smaller than the diffusivity of the cations in the bulk. In particular, for pore loadings larger than ρ_{bulk} , the axial diffusivity of the cations can be 1–2 orders of magnitude lower than the bulk diffusivity of [dmim⁺]. Another interesting observation in the inset of Figure 8 is that for the CMK-3 materials with 1.2 and 1.8 nm pore sizes there is a maximum in the diffusivity when the pore loading is equal to 0.9 ρ_{bulk} for the CMK-3 material with a pore size of 0.9 nm, the maximum in axial diffusivity is observed at a pore loading of 0.75 ρ_{bulk} . These peaks in the dynamics can be explained in terms of the variations in the axial density profiles of the ions at $\rho < \rho_{\text{bulk}}$ (Figure 4). These results suggest that at these low pore loadings the ions prefer to cluster together to create regions where the local density is similar to ρ_{bulk} , leaving regions where the density is significantly lower. These heterogeneities in the axial structure of the IL are likely to cause the nonmonotonic variations in dynamics with pore loading for a given pore size. In particular, the results shown in Figure 4 for a pore size of 1.2 nm indicate that the variations in axial density for a pore loading of 0.8 ρ_{bulk} are somewhat stronger than those observed for 0.9 ρ_{bulk} . For the former pore loading, the high-density regions have a larger average density and the

low-density regions have a larger size and exhibit a smaller average density, as compared to those observed for a pore loading of 0.9 ρ_{bulk} .

4. CONCLUDING REMARKS

We studied the structural and dynamical properties of the IL [dmim⁺][Cl[−]] confined inside a model CMK-3 nanoporous carbon material. Our results indicate that variables such as pore size, pore loading, and pore morphology have a profound influence on the structural and dynamical properties of the confined IL. The mass density profiles of the ions show evidence of significant layering, with the number of layers and the relative positions of the density peaks for [dmim⁺] and [Cl[−]] varying with pore size and pore loading. Our simulations also indicate that the cations close to the pore surface tend to align with their imidazolium ring parallel to the surfaces. In addition, confinement effects in a CMK-3 material are heterogeneous and significantly different from those observed in simple slit-like and cylindrical pores; the density profiles (Figure 2) show significant differences when determined along the straight-line and dashed-line paths indicated in Figure 1. Large heterogeneities in the density of the confined IL in the axial direction are observed at pore loadings below ρ_{bulk} (Figure 4), suggesting that the ions tend to cluster together in order to create regions where the local density is similar to ρ_{bulk} , leaving other regions of the pore partially depleted and exhibiting densities that are much lower than ρ_{bulk} . The radial distribution functions exhibit features similar to those observed for the $g(r)$ of the bulk IL; however, important variations in the height of the peaks are observed as the pore loading changes.

Regarding the dynamical properties, our results for the MSD, diffusivities, and single-particle time correlation functions for the rotational motion of the cations indicate that the dynamics of the confined [dmim⁺][Cl[−]] are significantly slower than those observed in bulk systems. For all pore sizes and pore loadings, it was found that the confined cations move faster than the anions. Our results indicate that the IL confined in CMK-3 materials exhibit dynamics in the z direction (free of confinement) that are similar in magnitude to the dynamics in the x and y directions (subject to confinement). These findings contrast to what was observed in a slit pore of similar size and are mainly due to the surface roughness of the carbon nanorods and the heterogeneous pore structure of CMK-3. Reductions in pore size cause the confined ions to spend a significantly larger amount of time in the ‘cage regime’ as compared to bulk systems. For pore loadings similar to ρ_{bulk} , the axial MSDs increase monotonically with increases in pore size, in contrast to what we found for [bmim⁺][PF₆[−]] inside carbon nanotubes, for which the axial MSD varied nonmonotonically with pore size.³⁰ For fixed pore sizes, the axial MSDs decrease monotonically as the pore loading increases above ρ_{bulk} but nonmonotonic variations in the axial MSDs are observed for pore loadings below ρ_{bulk} . Our results indicate that, for any given pore size, the axial diffusivity of the confined cations reach maxima at pore loadings below ρ_{bulk} (Figure 8), possibly due to the heterogeneities observed in the density in the axial direction at such low pore loadings (Figure 4). In upcoming studies we will consider ILs confined in other nanoporous materials with complex pore geometries as well as the effect of the presence of charges in the pore walls.

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